Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/06/2004
PI Name:	Hermanson, James C Ph.D.		
Project Title:	MODIFICATION OF TURBULENT NON-PREMIXED FLAMES BY PULSED FUEL INJECTION		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195-6605	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Performance Goal No.:			
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Task Description:	Fully-modulated fuel jets consisting of unheated gaseous hydrocarbon fuel (ethylene) are injected into a co-flowing or stagnant oxidizer. Optical diagnostics and probes are used to determine the flame length, the velocity field and the rate of oxidizer entrainment, the combustion temperatures, flame radiation, and exhaust emissions of pulsed and steady flames in both normal and reduced gravity. Experiments are conducted in the laboratory to characterize the baseline flame behavior at normal gravity conditions and in microgravity using the NASA GRC 2.2 second Drop Tower and the 5.18 s Zero-Gravity Facility. Laboratory experiments to examine the flame structure and exhaust emissions are performed at the UW; laboratory velocity field measurements are undertaken at WPI. The combustion characteristics of fully-modulated diffusion flames are also investigated by three-dimensional, unsteady computations using both direct numerical and large eddy simulation approaches. The overall objective of this research program is to determine the behavior of fully-modulated (i.e., fuel completely shut off during part of each pulsing cycle) and steady turbulent diffusion flames in both normal and reduced gravity. Emphasis is placed on examining the effects of buoyancy on the development and interaction of the turbulent flame structures over a wide range of injection conditions, ranging from isolated, non-interacting flame puffs to flames with steady injection. The implications of the dynamics and interaction of the burning fuel puffs for fuel/air mixing, combustion, and exhaust emissions are to be established. Combustion is critical to many systems essential to the technical infrastructure of the United States and of the world. Examples of important applications of combustion technology range from domestic furnaces and water heaters to large-scale central power stations. Turbulent combustion, which involves the complex coupling of chemical and fluid mechanical processes, is a critical fundamental problem for combustion processes. Specifically, combustion devices that are based on unsteady, turbulent flow appear to have the potential to achieve the low levels of emissions required to meet increasingly stringent environmental regulations, as well as economical savings that might derive from the use of more compact combustors. However, the physical phenomena responsible for the enhanced mixing in unsteady, reacting flows are not currently well understood (the mechanisms operative in steady, turbulent jet flames are also not completely understood). Part of the difficulty in either case stems from the presence of buoyancy, which significantly complicates the turbulent mixing in flames in normal gravity. By the use of the microgravity environment to effectively remove the effects of buoyancy, an improved fundamental understanding of the fuel/air mixing and combustion characteristics of both fully-modulated and steady-state combustion systems can result. The resulting new fundamental knowledge may then lead to improved combustor designs for higher efficiency, more economical operation, and lower emissions for numerous applications on earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work to date suggests that there are apparently specific injection conditions where the emissions of fully-modulated flames are comparable to those of the steady flame while the flame length is significantly shorter. This points to the potential utility of this technique in the development of compact, low emissions combustors involving turbulent diffusion flames.
Task Progress:	The combustor from an existing Drop Tower Rig has been installed and is operational in a new Combustion Laboratory at the UW. The combustor (10 x 10 cm cross section and 52 cm in length downstream of the nozzle tip) is supplied by ethylene fuel and house compressed air. Fully-modulated flames have been generated over the full range of pulsing conditions (injection times ranging from 6 ms to steady flames). A high framing rate CCD camera is operational and provides for imaging of the laboratory pulsed flames at up to 400 frames/second. High-speed images of the pulsed flames are currently being acquired. A pair of thermocouples with known separation has been configured to measure directly the celerity of flame puffs in normal gravity. A suite of emissions instruments is being assembled to measure CO, CO2, O2, and NOx concentrations; this includes an existing unburned hydrocarbons analyzer. In order to enhance the temporal resolution of the Particle Imaging Velocimetry (PIV) for velocity-field measurements, a reduced nozzle exit velocities and increased injection times are desired. The nozzle diameter in an existing combustor at WPI has been increased to 5 mm, resulting in a 2.5-fold decrease in the nozzle velocity compared with previous experiments. To accommodate the larger nozzle, the co-flow chamber has also been enlarged to a 30 x 30 cm square. An added benefit of the larger nozzle is that a much longer injection time (over 6-fold longer than previously) can be obtained for the same normalized puff injection volume used in previous experiments. The larger combustion chamber with glass walls, pulsing system, and fuel jet and air supplies has been assembled. Computations has been carried out to assess the Stokes number of various sized particles to be used in conjunction with the PIV system suggests and optimum particle size of 20 microns. To avoid clogging the small orifice pulsing valve, seed particles are injected into the combustion chamber co-flow. A cyclone seeder has been designed and constructed to introduce alumina particles into the air co-flow. The seeder can handle particle sizes in the range of 1 to 40 microns. The feasibility of performing emissions measurements in microgravity using the NASA GRC Zero-G Facility (ZGF) is being studied. Analysis indicates that, for most conditions, the ppm concentration levels of emissions in the entire ZGF chamber after one drop will not be sufficiently high to enable reliable measurements of the concentrations of CO and unburned hydrocarbons (UHC). Options for providing for a centerline-mounted, aspirated sampling probe, coupled with an initially evacuated vessel mounted external to the ZGF chamber are currently being evaluated.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Articles in Peer-reviewed Journals	Hermanson JC, Ghaem-Maghami E, Johari H. "CO/Unburned hydrocarbon emissions of strongly-pulsed, turbulent diffusion flames." Combustion Science and Technology, 2004;176(11):1855-66. http://dx.doi.org/10.1080/00102200490504481 , Oct-2004
Articles in Peer-reviewed Journals	Hermanson JC, Johari H, Stocker DP, Hegde UG. "Buoyancy effects in strongly pulsed turbulent diffusion flames." Combustion and Flame, 2004 Oct;139(1-2):61-76. http://dx.doi.org/10.1016/j.combustflame.2004.08.005 , Oct-2004

Papers from Meeting Proceedings	Hermanson JC, Johari H, Stocker DP, Hegde UG. "Thermal Characteristics and Structure of Fully-Modulated, Turbulent Diffusion Flames in Microgravity." 42nd AIAA Aerospace Sciences Meeting, Reno, NV, January 5-8, 2004. 42nd AIAA Aerospace Sciences Meeting, Reno, NV, January 5-8, 2004. AIAA Paper 2004-0959 , Jan-2004
Presentation	Hermanson, J.C., Johari, H., Stocker, D.P. and Hegde, U.G. "Effects of Buoyancy on the Structure and Combustion Temperatures of Strongly-Pulsed, Turbulent Diffusion Flam" Thirtieth International Symposium on Combustion, Chicago, IL, (poster presentation) Jul-2004
Presentation	Hermanson, J.C., Ghaem-Maghami, E. and Johari, H. "Exhaust Emissions of Strongly-Pulsed, Turbulent Diffusion Flames" Thirtieth International Symposium on Combustion, Chicago, IL, (poster presentation) Jul-2004
Presentation	Hermanson, J.C., Johari, H., Stocker, D.P. and Hegde, U.G. "Buoyancy Effects in Strongly-Pulsed, Turbulent Diffusion Flames" Strategic Research to Enable NASA’s Exploration Missions, Cleveland, OH, (poster presentation) Jun-2004
Presentation	Hermanson, J.C. "Buoyancy Effects and Emissions in Strongly-Pulsed, Turbulent Diffusion Flames" University of California at San Diego Oct-2004